Probleme IS

Seria AA

1) 3 bănci

$p_1 = 0,8$

$p_2 = 0,75$

$p_3 = 0,82$

A = evenimentul „2 răspunsuri favorabile și unul nefavorabil"

Utilizăm schema lui Poisson.

P(A) va fi coeficientul lui x^2 din dezvoltarea $\prod_{i=1}^{3}(p_i x + q_i)$, $q_i = 1 - p_i$

$$\begin{cases} q_1 = 0,2 \\ q_2 = 0,25 \\ q_3 = 0,18 \end{cases} \Rightarrow \prod_{i=1}^{3}(p_i x + q_i) =$$

$$= (0,8x + 0,2)(0,75x + 0,25)(0,82x + 0,18) =$$

$$= 0,8 \cdot 0,75 \cdot 0,18 + 0,2 \cdot 0,75 \cdot 0,82 + 0,8 \cdot 0,25 \cdot 0,82 =$$

$$= 0,395$$

Cutia 1: 6 verzi, 3 galbene

Cutia 2: 5 verzi, 4 galbene

(2.) A = evenimentul „balonul este galben"

B_i = evenimentul „balonul provine din cutia i^4, $i \in \{1,2\}$

a) Conform formulei probabilității totale:

(B_1 și B_2 formează un sistem complet de evenimente)

$$P(A) = P(A/B_1) \cdot P(B_1) + P(A/B_2) \cdot P(B_2)$$

$$P(B_1) = \frac{9}{18} = \frac{1}{2} \qquad P(B_2) = \frac{9}{18} = \frac{1}{2}$$

$$P(A/B_1) = \frac{3}{9} = \frac{1}{3}$$

$$P(A/B_2) = \frac{4}{9}$$

$$\Rightarrow P(A) = \frac{1}{2} \cdot \frac{1}{3} + \frac{4}{9} \cdot \frac{1}{2} = \frac{1}{6} + \frac{4}{18} = \frac{3+4}{18} = \frac{7}{18}$$

b) $P(B_2/A)$ = probabilitatea ca balonul să fie din cea de a doua cutie, știind că acesta este galben

$$P(B_2/A) = \frac{P(B_2 \cap A)}{P(A)}$$

$$P(A/B_2) = \frac{P(B_2 \cap A)}{P(B_2)} \Rightarrow$$

$$P(B_2 \cap A) = P(A/B_2) \cdot P(B_2)$$

$$P(B_2 \cap A) = \frac{1}{2} \cdot \frac{4}{9} = \frac{2}{9}$$

$$A = \frac{7}{18} \Rightarrow$$

$$P(B_2/A) = \frac{\frac{2}{9}}{\frac{7}{18}} = \frac{2}{7} \cdot 2 = \frac{4}{7}$$

(2.) $X \sim U[1,2]$

$(\sqrt{y})' = (y^{\frac{1}{2}})' = \frac{1}{2} \cdot \frac{1}{\sqrt{y}} = \frac{1}{2\sqrt{y}}$

$P(2 < x^2 < 5)$

Densitatea de probabilitate a lui X este:

$$f_X(x) = \begin{cases} \frac{1}{2-1}, & 1 \le x \le 2 \\ 0, & \text{in rest} \end{cases} = \begin{cases} 1, & 1 \le x \le 2 \\ 0, & \text{in rest} \end{cases}$$

Funcția de repartiție a lui $Y = x^2$ este

$$F_Y(y) = P(Y < y) = P(|X| < \sqrt{y}) =$$

$$= P(-\sqrt{y} < x < \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$\Rightarrow f_Y(y) = F_Y'(y) = F_X'(\sqrt{y}) - F_X'(-\sqrt{y}) =$$

$$= f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} - f_X(-\sqrt{y}) \cdot \left(-\frac{1}{2\sqrt{y}}\right) =$$

$$= \frac{1}{2\sqrt{y}} \cdot \left(f_X(\sqrt{y}) + f_X(-\sqrt{y})\right) =$$

$$= \begin{cases} \frac{1}{2\sqrt{y}}, & y \in [1,4] \\ 0, & \text{in rest} \end{cases} \Rightarrow$$

$$P(2 < x^2 < 5) = \int_2^5 f_Y(y)\,dy = \int_2^4 f_Y(y)\,dy =$$

$$= \frac{1}{2}\int_2^4 \frac{1}{\sqrt{y}}\,dy = \frac{1}{2}\cdot\int_2^4 y^{-\frac{1}{2}}\,dy = \left(\cancel{\frac{1}{2}}\cdot y^{\frac{1}{2}}\cdot\frac{1}{\cancel{\frac{1}{2}}}\right)\Big|_2^4$$

$$= \sqrt{y}\Big|_2^4 = 2-\sqrt{2}$$

4. $X \sim Exp(\lambda_1 = \frac{1}{30})$

$Y \sim Exp(\lambda_2 = \frac{1}{10})$

$P(X+Y>30)$ = probabilitatea ca tranzactia să dureze mai mult de o jumătate de oră

$$f_X(x) = \begin{cases} \frac{1}{30}e^{-x/30}, & x>0 \\ 0, & \text{în rest} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{10}e^{-y/10}, & y>0 \\ 0, & \text{în rest} \end{cases}$$

X, Y v.a. independente $\Rightarrow$

$$f(x,y) = f_X(x)\cdot f_Y(y) = \begin{cases} \frac{1}{300}\cdot e^{-\left(\frac{x}{30}+\frac{y}{10}\right)}, & x>0,\ y>0 \\ 0, & \text{în rest} \end{cases}$$

↓

densitatea de probabilitate a vectorului (X,Y)

$$P(X+Y)>30 = 1 - P(X+Y \leq 30) =$$

$$= 1 - \int_0^{30} \int_0^{30-x} \frac{1}{300} \cdot e^{-\left(\frac{x}{30}+\frac{y}{10}\right)} \, dy \, dx =$$

$$= 1 - \int_0^{30} -\frac{1}{30} \left(e^{\frac{-x+45}{15}} - e^{-\frac{x}{30}} \right) dx =$$

$$= 1 - \left[\frac{-3e^2-1}{2e^3} + 1 \right] = \frac{-3e^2-1}{2e^3} \simeq 0{,}53$$

⑤ $$f(x,y) = \begin{cases} \frac{xy^2+x^2y}{4}, & (x,y) \in [0,1] \times [2,3] \\ 0, & \text{în rest} \end{cases}$$

$\rho(X,Y)$ = coeficient de corelație

$$\rho(X,Y) = \frac{cov(X,Y)}{\sqrt{D^2(X) \cdot D^2(Y)}}$$

$$cov(X,Y) = E(XY) - E(X) \cdot E(Y)$$

$$f_X(x) = \int_0^{\infty} f(x,y) \, dy = \begin{cases} \int_2^3 \frac{xy^2+x^2y}{4} \, dy, & x \in [0,1] \\ 0, & \text{în rest} \end{cases}$$

$$= \begin{cases} \frac{1}{4}\left(\frac{5}{2}x^2 + \frac{19}{3}x\right), & x \in [0,1] \\ 0, & \text{in rest} \end{cases}$$

$$f_Y(y) = \int_0^{\infty} f(x,y)\,dx = \begin{cases} \int_0^1 \frac{xy^2 + x^2y}{4}\,dx, & y \in [2,3] \\ 0, & \text{in rest} \end{cases}$$

$$= \begin{cases} \frac{1}{4}\left(\frac{1}{2}y^2 + \frac{1}{3}y\right), & y \in [2,3] \\ 0, & \text{in rest} \end{cases}$$

$$E(X) = \int_{-\infty}^{+\infty} x \cdot f_X(x)\,dx = \int_0^1 x \cdot \frac{1}{4}\left(\frac{5}{2}x^2 + \frac{19}{3}x\right)dx =$$

$$= \frac{197}{288}$$

$$E(Y) = \int_{-\infty}^{+\infty} y \cdot f_Y(y)\,dy = \int_2^3 y \cdot \frac{1}{4}\left(\frac{1}{2}y^2 + \frac{1}{3}y\right)dy =$$

$$= \frac{737}{288}$$

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 \cdot f_X(x)\,dx = \int_0^1 x^2 \cdot \frac{1}{4}\left(\frac{5}{2}x^2 + \frac{19}{3}x\right)dx$$

$$= \frac{25}{48}$$

$$E(Y^2) = \int_{-\infty}^{+\infty} y^2 \cdot f_Y(y)\, dy = \int_2^3 y^2 \cdot \frac{1}{4}\left(\frac{1}{2}y^2 + \frac{1}{3}y\right) dy =$$

$$= \frac{1591}{240}$$

$$D^2(X) = E(X^2) - (E(X))^2 = \frac{25}{48} - \left(\frac{197}{288}\right)^2 \simeq 0{,}0529\,39$$

$$D^2(Y) = E(Y^2) - (E(Y))^2 = \frac{1591}{240} - \left(\frac{737}{288}\right)^2 \simeq 0{,}08054$$

$$E(XY) = \int_0^1 \int_2^3 xy\, f(x,y)\, dy\, dx =$$

$$= \int_0^1 \int_2^3 xy \cdot \frac{xy^2 + x^2 y}{4}\, dy\, dx = \frac{7}{4}$$

$$\Rightarrow \quad cov(X,Y) = E(XY) - E(X) \cdot E(Y) =$$

$$= \frac{7}{4} - \frac{197}{288} \cdot \frac{737}{288} \simeq -0{,}00074460$$

$$\rho(X,Y) = \frac{cov(X,Y)}{\sqrt{D^2(X) \cdot D^2(Y)}} = \boxed{-0{,}00683}$$

(6.) Şir de variabile aleatoare independente repartizate Poisson de parametru $\lambda = 2$

$$Y_{100} = \sum_{i=1}^{100} X_i$$

$$P(190 < Y_{100} < 210) = ?$$

$E(X_i) = \lambda = 2$

$D^2(X_i) = \lambda = 2$

$i = \overline{1,100}$

→ dispersie + media pt. variabile aleatoare repartizate Poisson

$\Rightarrow E(Y_{100}) = 100 \cdot 2 = 200$

$D^2(Y_{100}) = 100 \cdot 2 = 200$

$\Rightarrow P(190 < Y_{100} < 210) = P\left(\frac{190 - 200}{\sqrt{200}} < \frac{Y_{100} - 200}{\sqrt{200}} < \frac{210 - 200}{\sqrt{200}} \right) = P(-0,707 < \frac{Y_{100} - 200}{\sqrt{200}} < 0,707) = \Phi(0,707) - \Phi(-0,707) =$

$= \Phi(0,707) - (1 - \Phi(0,707)) = 2 \cdot \Phi(0,707) - 1 =$

$\simeq 2 \cdot \Phi(0,71) - 1 = 2 \cdot 0,7611 - 1 = 0,5222$

Varianta 2 – seria AA

1) Folosim schema Poisson.

A_i = evenimentul „figiderul din lotul i selectat este defect", $i = \overline{1,3}$

$P(A_1) = p_1 = 0{,}04$ $\quad P(\overline{A_1}) = q_1 = 0{,}96$

$P(A_2) = p_2 = 0{,}02$ $\quad P(\overline{A_2}) = q_2 = 098$

$P(A_3) = p_3 = 0{,}03$ $\quad P(\overline{A_3}) = q_3 = 0{,}97$

Probabilitatea ca 2 figidere să fie defecte și unul bun este coeficientul lui x^2 din dezvoltarea $\prod_{i=1}^{3} (p_i \cdot x + q_i)$, $q_i = 1 - p_i$

$$\prod_{i=1}^{3} (p_i \cdot x + q_i) = (p_1 \cdot x + q_1)(p_2 \cdot x + q_2) \cdot$$

$$(p_3 \cdot x + q_3) = p_1 \cdot p_2 \cdot q_3 + p_1 \cdot q_2 \cdot p_3 + q_1 \cdot p_2 \cdot p_3$$

$$= 0{,}04 \cdot 0{,}02 \cdot 0{,}97 + 0{,}04 \cdot 098 \cdot 0{,}03 +$$

$$096 \cdot 0{,}02 \cdot 0{,}03 = 0{,}002528$$

Seria AB

1. 1000 piese $\longrightarrow$ 2,5% defecte

$a = \frac{2,5}{100} \cdot 1000 = 2,5 \cdot 10 = 25$ piese defecte

$b = 1000 - 25 = 975$ piese ok

Aplicăm schema hipergeometrică.

$A =$ evenimentul „5 piese sunt defecte"

$$P(A) = \frac{C_{25}^{5} \cdot C_{975}^{85}}{C_{1000}^{90}}$$

$B_i =$ evenimentul „i piese sunt defecte"

$B =$ „88 piese cel puțin sunt bune"

$P(B) = P(B_2) + P(B_1) + P(B_0) =$

$$= \frac{C_{25}^{2} \cdot C_{975}^{88}}{C_{1000}^{90}} + \frac{C_{25}^{1} \cdot C_{975}^{89}}{C_{1000}^{90}} + \frac{C_{25}^{0} \cdot C_{975}^{90}}{C_{1000}^{90}}$$

(2.) $f(x)=\begin{cases} ke^{-\alpha x}(1-e^{-\alpha x}), & x\geq 0, \alpha>0 \\ 0, \text{ in rest} \end{cases}$

a) constanta $k=?$ a.î. f este densitate de probabilitate a unei v.a. X

Punem condiția:

$$i=\int_{-\infty}^{\infty} f(x)\,dx=1 \iff \int_{0}^{+\infty} k\cdot e^{-\alpha x}(1-e^{-\alpha x})\,dx=$$

$$=k\int_{0}^{+\infty} e^{-\alpha x}\,dx - k\int_{0}^{+\infty} e^{-2\alpha x}\,dx$$

Pentru prima integrală facem schimbarea de variabilă $t=\alpha x$ și pt. a doua integrală $t=2\alpha x \Rightarrow$

$$\begin{cases} t=\alpha x \Rightarrow dt=\alpha\,dx \\ t=2\alpha x \Rightarrow dt=2\alpha\,dx \end{cases} \Rightarrow$$

$$i=k\int_{0}^{+\infty} e^{-t}\cdot t^{0}\cdot\frac{1}{\alpha}\,dt - k\int_{0}^{\infty} e^{-t}\cdot t^{0}\cdot\frac{1}{2\alpha}\,dt$$

$$=\frac{k}{\alpha}\cdot\Gamma(1)-\frac{k}{2\alpha}\Gamma(1)=\frac{k}{2\alpha}$$ (am folosit funcția gama a lui Euler)

Cum $\int_{-\infty}^{+\infty} f(x)\,dx=1 \Rightarrow \frac{k}{2\alpha}=1 \Rightarrow k=2\alpha$

Pt. f se respectă și prop. $f(x) \geq 0$,

deoarece $k > 0$ și $1 - \underbrace{e^{-\alpha x}}_{<1} > 0$, $e^{-\alpha x} > 0$.

b) Funcția de repartiție a variabilei aleatoare X.

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt = \int_{0}^{x} 2\alpha \cdot e^{-\alpha t}(1 - e^{-\alpha t})\,dt =$$

$$= \int_{0}^{x} 2\alpha \cdot e^{-\alpha t}\,dt - \int_{0}^{x} 2\alpha \cdot e^{-2\alpha t}\,dt =$$

$$= -2e^{-\alpha t}\Big|_0^x - \left(-e^{-\alpha t}\right)\Big|_0^x =$$

$$= -2 \cdot e^{-\alpha x} + 2 + e^{-\alpha x} - 1 = 1 - e^{-\alpha x}$$

$$\Rightarrow F_X(x) = \begin{cases} 1 - e^{-\alpha x}, & x \geq 0,\ \alpha > 0 \\ 0, & \text{în rest} \end{cases}$$

c) $$E(X) = \int_{-\infty}^{+\infty} x f(x)\,dx =$$

$$= 2\alpha \int_{0}^{+\infty} x e^{-\alpha x}(1 - e^{-\alpha x})\,dx =$$

$$= 2\alpha \left(\int_{0}^{+\infty} x \cdot e^{-\alpha x}\,dx - \int_{0}^{+\infty} x \cdot e^{-2\alpha x}\,dx \right)$$

Facem schimbările de variabilă:

$$\begin{cases} t = \alpha x \iff dt = \alpha\, dx \\ t = 2\alpha x \iff dt = 2\alpha\, dx \end{cases}$$

$$\Rightarrow E(x) = 2\alpha \cdot \left(\int_0^{+\infty} \frac{1}{\alpha} \cdot t \cdot e^{-t} \cdot \frac{1}{\alpha}\, dt - \int_0^{+\infty} \frac{1}{2\alpha} \cdot t \cdot e^{-t} \cdot \frac{1}{2\alpha}\, dt \right) =$$

$$= \frac{1}{\alpha} \left(2 \int_0^{+\infty} t^{2-1} \cdot e^{-t}\, dt - \frac{1}{2} \int_0^{+\infty} t^{2-1} \cdot e^{-t}\, dt \right) =$$

$$= \frac{1}{\alpha} \left(2\Gamma(2) - \frac{1}{2}\Gamma(2) \right) = \frac{1}{\alpha} \cdot \frac{3}{2} \cdot \Gamma(2) = \frac{3}{2} \cdot \alpha$$

(3.) $X \sim Exp(\lambda_1 = 3)$

$Y \sim Exp(\lambda_2 = 5)$

$U = 3X + 5Y$

$V = \frac{X}{Y}$

$(U, V) = (3X + 5Y, \frac{X}{Y})$

$f_{(U,V)}(u,v) = ?$

U, V independente ?

Facem schimbarea de variabile

$u = 3x + 5y$ si $v = \frac{x}{y} \Rightarrow$

$x = vy$

$u = 3vy + 5y \Leftrightarrow y(3v+5) = u \Leftrightarrow y = \frac{u}{3v+5}$

$x = vy = \frac{uv}{3v+5}$

$$\frac{D(x,y)}{D(u,v)} = \begin{vmatrix} \frac{v}{3v+5} & \frac{5u}{(3v+5)^2} \\ \frac{1}{3v+5} & \frac{-3u}{(3v+5)^2} \end{vmatrix} =$$

$$= \frac{-3uv}{(3v+5)^3} - \frac{5u}{(3v+5)^3} = \frac{-u(3v+5)}{(3v+5)^3} =$$

$$= \frac{-u}{(3v+5)^2}$$

$$\begin{cases} X \sim Exp(\lambda_1 = 3) \\ Y \sim Exp(\lambda_2 = 5) \end{cases} \Rightarrow$$

$$f_X(x) = \begin{cases} 3e^{-3x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$f_Y(y) = \begin{cases} 5e^{-5y}, & y>0 \\ 0, & y \le 0 \end{cases}$$

X, Y independente $\Rightarrow$

$$f_{(X,Y)}(x,y) = f_X(x) \cdot f_Y(y) =$$

$$= \begin{cases} 15\, e^{-(3x+5y)}, & x>0, y>0 \\ 0, & \text{in rest} \end{cases}$$

obținem $f_{(U,V)}(u,v) =$

$$\begin{cases} 15\, e^{-\left(3\cdot \frac{uv}{3v+5} + 5\cdot \frac{u}{3v+5}\right)} \cdot |J|, & \frac{uv}{3v+5}>0, \frac{u}{3v+5}>0. \\ 0, & \text{in rest} \end{cases}$$

$$= \begin{cases} 15 \cdot e^{-\frac{u(3v+5)}{3v+5}} \cdot \frac{u}{(3v+5)^2}, & u,v>0 \\ 0, & \text{in rest} \end{cases} =$$

$$= \begin{cases} 15 \cdot e^{-u} \cdot \frac{u}{(3v+5)^2}, & u,v>0 \\ 0, & \text{in rest} \end{cases}$$

$$f_U(u) = \int_0^{+\infty} 15 \cdot e^{-u} \cdot \frac{u}{(3v+5)^2}\, dv =$$

$$= 15 \cdot e^{-u} \cdot u \cdot \left(-\frac{1}{9v+15}\right)\Big|_0^{+\infty} = 15 \cdot \frac{1}{15} \cdot e^{-u} \cdot u =$$

$$= e^{-u} \cdot u, \quad u \in (0,+\infty), \quad 0 \text{ în rest.}$$

$$f_V(v) = \int_0^{+\infty} 15 \cdot e^{-u} \cdot \frac{u}{(3v+5)^2} \cdot du = 15 \cdot \frac{1}{(3v+5)^2} \cdot \underbrace{\Gamma(2)}_{=1}$$

$= 15 \cdot \frac{1}{(3v+5)^2}$ (am folosit funcția gama a lui Euler), $v \in [0,+\infty]$, 0 în rest

$$\Rightarrow f_U(u) \cdot f_V(v) = f_{(U,V)}(u,v) \Rightarrow$$

U, V sunt independente.

(4.) X = mărimea semnalului emis

$Y = \alpha X + \delta \longrightarrow$ perturbări

↳ coeficient de amplificare

δ și X independente

$E(X) = m \qquad E(\delta) = 0$

$D^2(X) = 1 \qquad D^2(\delta) = \sigma^2$

$$\rho(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sqrt{D^2(X)\,D^2(Y)}}$$

$$\operatorname{cov}(X,Y) = E(XY) - E(X)\cdot E(Y)$$

$$E(Y) = E(\alpha X + \delta) = E(\alpha X) + E(\delta) =$$

$$= \alpha E(X) + E(\delta) = \alpha m + 0 = \alpha m$$

$$E(XY) = E(X(\alpha X + \delta)) = E(\alpha X^2 + X\delta) =$$

$$= E(\alpha X^2) + E(X\delta) \overset{X,\delta}{\underset{\text{independente}}{=\!=\!=}} \alpha E(X^2) +$$

$$E(X)\cdot \underbrace{E(\delta)}_{=0} = \alpha E(X^2)$$

$D^2(X) = 1 \Rightarrow \quad E(X^2) - (E(X))^2 = 1 \Rightarrow$

$E(X^2) = 1 + (E(X))^2 = 1 + m^2 \Rightarrow$

$E(XY) = \alpha(1+m^2)$

$\Rightarrow \operatorname{cov}(X,Y) = E(XY) - E(X)\cdot E(Y) =$

$= \alpha(1+m^2) - m\cdot\alpha m = \alpha + \alpha m^2 - m^2\alpha = \alpha$

$D^2(X) = 1$

$D^2(Y) = E(Y^2) - (E(Y))^2$

$E(Y^2) = E((X\cdot\alpha + \delta)^2) = E(X^2\cdot\alpha^2 + 2\alpha X\delta + \delta^2)$

$= \alpha^2 E(X^2) + 2\alpha E(X\delta) + E(\delta^2) =$

$= \alpha^2(1+m^2) + 2\alpha E(X)\underbrace{E(\delta)}_{=0} + E(\delta^2) =$

$= \alpha^2(1+m^2) + E(\delta^2)$

$D^2(\delta) = \sigma^2 \iff E(\delta^2) - \overbrace{(E(\delta))^2}^{=0} = \sigma^2 \Rightarrow$

$E(\delta^2) = \sigma^2 \Rightarrow$

$E(Y^2) = \alpha^2(1+m^2) + \sigma^2 \Rightarrow$

$D^2(Y) = E(Y^2) - (E(Y))^2 = \alpha^2(1+m^2) + \sigma^2 - \alpha^2 m^2$

$= \alpha^2 + \sigma^2 \Rightarrow$

$$\rho(X,Y) = \frac{\text{cov}(X,Y)}{\sqrt{D^2(X) \cdot D^2(Y)}} = \frac{\alpha}{\sqrt{1 \cdot (\alpha^2+\sigma^2)}} =$$

$$= \frac{\alpha}{\sqrt{\alpha^2+\sigma^2}} < 1$$

5. posibili cumpărători $\Rightarrow n = 1500$

$p = 0{,}85 \Rightarrow q = 0{,}15$

$X =$ nr. de cumpărători

Folosim formula Moivre Laplace

$P(1260 < X < 1290) =$

$$= P\left(\frac{1260 - 1500 \cdot 0{,}85}{\sqrt{1500 \cdot 0{,}85 \cdot 0{,}15}} < \frac{X - np}{\sqrt{npq}} < \right.$$

$$\left. \frac{1290 - 1500 \cdot 0{,}85}{\sqrt{1500 \cdot 0{,}85 \cdot 0{,}15}} \right) =$$

$$= P\left(-1{,}08 < \frac{x-np}{\sqrt{npq}} < 1{,}08\right) =$$

$$= \Phi(1{,}08) - \Phi(-1{,}08) = 2\Phi(1{,}08) - 1 =$$

$$= 2 \cdot 0{,}8599 - 1 = 0{,}7198$$

(6.) $P = \begin{pmatrix} 0{,}45 & ? \\ 0{,}65 & ? \end{pmatrix} \Rightarrow$ De refăcut Markov!

$$P = \begin{pmatrix} 0{,}45 & 0{,}55 \\ 0{,}65 & 0{,}35 \end{pmatrix}$$

$$\pi_0 = \left(\frac{3}{10}, \frac{7}{10}\right)$$

a) Matricea de trecere după 2 pași este

P(2) $P^2 = \begin{pmatrix} 0{,}45 & 0{,}55 \\ 0{,}65 & 0{,}35 \end{pmatrix} \begin{pmatrix} 0{,}45 & 0{,}55 \\ 0{,}65 & 0{,}35 \end{pmatrix} =$

$$= \begin{pmatrix} \frac{14}{25} & \frac{11}{25} \\ \frac{13}{25} & \frac{12}{25} \end{pmatrix}$$

Lanțul este mai probabil să se afle în starea (1)

Pe termen lung:

Determinăm repartiția staționară a procesului

$$(p_1 \quad p_2) \begin{pmatrix} 0{,}45 & 0{,}55 \\ 0{,}65 & 0{,}35 \end{pmatrix} = (p_1 \quad p_2)$$

$$\begin{cases} 0{,}45 p_1 + 0{,}65 p_2 = p_1 \\ 0{,}55 p_1 + 0{,}35 p_2 = p_2 \\ p_1 + p_2 = 1 \end{cases} \Rightarrow$$

$$\begin{cases} -0{,}55 p_1 + 0{,}65 p_2 = 0 \\ p_1 + p_2 = 1 \end{cases} \Bigg| \begin{matrix} \cdot 100 \\ \Rightarrow \\ \cdot 55 \end{matrix}$$

$$\begin{cases} -55 p_1 + 65 p_2 = 0 \\ 55 p_1 + 55 p_2 = 55 \end{cases} \Rightarrow 120 p_2 = 55 \Rightarrow$$

$$\begin{cases} p_2 = \frac{11}{24} \\ p_1 = \frac{13}{24} \end{cases} \Rightarrow$$ pe termen lung matricea P tinde la

$$\begin{pmatrix} 13/24 & 11/24 \\ 13/24 & 11/24 \end{pmatrix}$$

Lanțul este mai probabil să se afle în starea 1.

$$\left(\lim_{n \to +\infty} p(n) = \pi = (13/24 \quad 11/24) \right)$$

b) $X_0 \sim \begin{pmatrix} 1 & 2 \\ 3/10 & 7/10 \end{pmatrix}$ $\quad p(0) = \left(\frac{3}{10} \quad \frac{7}{10}\right)$

$$p(2) = p(0) \cdot P^2 = (0{,}3 \quad 0{,}7) \cdot \begin{pmatrix} 14/25 & 11/25 \\ 13/25 & 12/25 \end{pmatrix}$$

$$= (133/250 \quad 117/250) \rightarrow$$

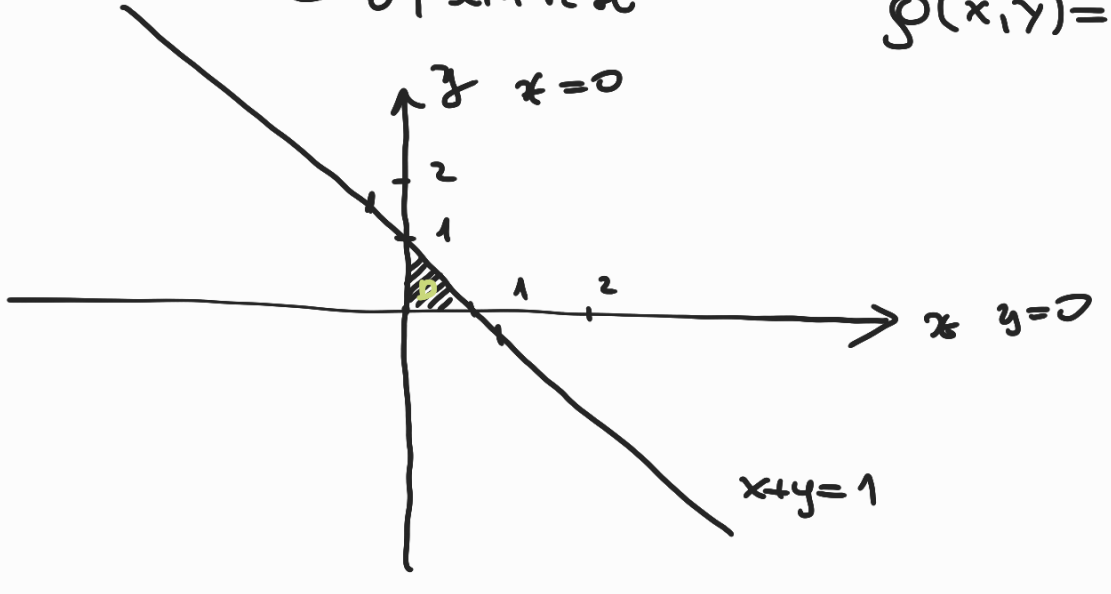
$$X_2 \sim \begin{pmatrix} 1 & 2 \\ 133/250 & 117/250 \end{pmatrix}$$

$$P(X_0 = 2 \mid X_2 = 1) = p_{12}(2) = \frac{11}{25}$$

(5)

$$f(x,y) = \begin{cases} 24xy, & (x,y) \in D \\ 0, & \text{im rest} \end{cases}$$

$g(x,y) = ?$



$$f_X(x) = \begin{cases} \int_0^{1-x} 24xy\,dy, & x \in [0,1] \\ 0, & \text{in rest} \end{cases} =$$

$$= \begin{cases} 12xy^2 \Big|_0^{1-x}, & x \in [0,1] \\ 0, & \text{in rest} \end{cases} =$$

$$= \begin{cases} 12x \cdot (1-x)^2, & x \in [0,1] \\ 0, & \text{in rest} \end{cases}$$

$$f_Y(y) = \begin{cases} \int_0^{1-y} 24xy\,dx, & y \in [0,1] \\ 0, & \text{in rest} \end{cases} =$$

$$= \begin{cases} 12x^2y \Big|_0^{1-y}, & y \in [0,1] \\ 0, & \text{in rest} \end{cases} =$$

$$= \begin{cases} 12(1-y)^2 y, & y \in [0,1] \\ 0, & \text{in rest} \end{cases}$$

$$\rho(X,Y)=\frac{\operatorname{cov}(X,Y)}{\sqrt{D^2(X)D^2(Y)}} \qquad \operatorname{cov}(X,Y)=E(XY)-E(X)E(Y)$$

$$E(X)=\int_{-\infty}^{\infty} x f(x)\,dx=\int_0^1 12x^2(1-x)^2\,dx=$$

$$=12\int_0^1 x^{3-1}\cdot(1-x)^{3-1}\,dx=12\,B(3,3)=$$

$$=12\cdot\frac{\Gamma(3)\cdot\Gamma(3)}{\Gamma(6)}=12\cdot\frac{2!\cdot 2!}{5!}=\frac{12\cdot 4\,2}{120}=\frac{2}{5}$$

$$E(Y)=\int_{-\infty}^{+\infty} y f(y)\,dy=\int_0^1 12y^2(1-y)^2\,dy=\text{analog}=\frac{2}{5}$$

$$E(XY)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xy\cdot f(x,y)\,dx\,dy=$$

$$=\iint_D (xy)^2\cdot 24\,dx\,dy=24\int_0^1\int_0^{1-x}(xy)^2\,dy\,dx=$$

$$=24\int_0^1 \frac{y^3}{3}\cdot x^2\Big|_0^{1-x}dx=8\int_0^1(1-x)^3\cdot x^2\,dx=$$

$$=8\cdot\int_0^1(1-x)^{4-1}\cdot x^{3-1}\,dx=8\cdot B(3,4)=$$

$$=8\cdot\frac{\Gamma(3)\cdot\Gamma(4)}{\Gamma(7)}=8\cdot\frac{2!\cdot 3!}{6!}=$$

$$= \frac{8^{2} \cdot 2 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \frac{2}{15}$$

$$\Rightarrow \quad \operatorname{cov}(X,Y) = E(XY) - E(X)E(Y) =$$

$$= \frac{2}{15} - \left(\frac{2}{5}\right)^2 = \frac{2}{15} - \frac{4}{25} = \frac{10}{75} - \frac{12}{75} = \frac{-2}{75}$$

$$E(x^2) = \int_{-\infty}^{+\infty} x^2 \cdot f_X(x)\,dx = \int_0^1 12\,x^3(1-x)^2\,dx$$

$$= 12 \int_0^1 x^{4-1} \cdot (1-x)^{3-1}\,dx = 12\,B(4,3) -$$

$$= 12 \cdot \frac{\Gamma(4)\cdot\Gamma(3)}{\Gamma(7)} = 12 \cdot \frac{2! \cdot 3!}{6!} = \frac{12 \cdot 2 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \frac{1}{5}$$

Analog $E(Y^2) = \frac{1}{5}$

$$\Rightarrow \quad D^2(X) = E(X^2) - (E(X))^2 = \frac{1}{5} - \left(\frac{2}{5}\right)^2 =$$

$$= \frac{1}{5} - \frac{4}{25} = \frac{5-4}{25} = \frac{1}{25}$$

Analog $D^2(Y) = \frac{1}{25} \Rightarrow$

$$\rho(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sqrt{D^2(X) \cdot D^2(Y)}} = \frac{\frac{-2}{75}}{\sqrt{\frac{1}{25} \cdot \frac{1}{25}}} =$$

$$= \frac{\frac{-2}{75}}{\frac{1}{5} \cdot \frac{1}{5}} = \frac{-2}{3}$$